

Predicting Factor Decay: ML Models for Cross-Factor Predictability Erosion

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Can we forecast when a cross-factor predictive relationship will erode, rather than only explain its disappearance afterward?

CENTRAL IDEA

Combine regime-conditional Granger diagnostics, a survival-style model of signal decay and online regime classification to study prospective monitoring of changing factor relationships.

MAIN FINDING

The submitted five-page paper reports a random-survival-forest C-index of 0.933 with five-fold standard deviation 0.097. The decay analysis has only seven events. Rolling HMM detection has a 6.6-day average delay with 47.6% classification accuracy, making the speed-accuracy tradeoff material.

EVIDENCE AND SCOPE

The submitted analysis considers 30 directed Fama-French factor pairs across five markets and reports state heterogeneity in 19 of 30 pairs. Initial signal strength is the leading reported decay predictor. These are submission claims, not results newly reproduced during repository extraction.

LIMITS

The primary US HML-to-SMB signal fails to replicate in the post-break out-of-sample period. Small event counts, specification search and modest economic effects make the evidence exploratory. It does not establish profitable trading or structural causality.

TAKEAWAY

Signal durability can be posed as a forecasting problem, but attractive discrimination metrics need to be interpreted alongside event counts, replication and operational error rates.

한국어 핵심 요약

요인 간 예측 관계가 사라지는 이유를 사후 설명하는 데서 나아가, 언제 약해질지 예측하려는 연구입니다. 제출본의 높은 C-index는 감쇠 사건이 7개뿐이라는 조건과 함께 읽어야 합니다. 주요 미국 신호의 표본 외 재현 실패와 탐지 속도·정확도 절충을 명시합니다. ICAIF 심사 결과는 아직 나오지 않았습니다.

Source: actual five-page submission; SHA-256 798939cc75e3f690...

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